CLAIMPROMOTEBENCH: PAIRED REGRESSION TESTS FOR VERSIONED AGENT POLICIES

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ABSTRACT

Repository policies are versioned inputs to coding agents, yet evaluations usually score versions independently. We present CLAIMPROMOTEBENCH, a paired regression protocol that binds two policy versions, an unchanged case, and a target action relation. For finite executable policies, it enumerates action-changing contexts and derives cardinality-minimum suites with relative rule and transition coverage. For prose policies, targets are declared author mappings, not policy-owner ground truth. Route records retain transport, syntax, schema, and semantic failures in end-to-end denominators. Eight public histories illustrate the supported fault classes. On ten author-mapped pairs from two `AGENTS.md` revisions, eight hosted routes yield 128/160 item agreements and 52/80 pair agreements. In a prespecified non-`ALLOW` extension, every valid item agrees with its author target; incomplete outputs leave 50/54 stable pairs but only 6/18 changed pairs complete. On selected routes, a same-item request-cap comparison changes complete target-agreeing pairs from 0/15 to 7/15 and 0/15 to 9/15, with one call per cell. The protocol supports auditable frozen-case regression tests. The executable studies also retain exact action confusion matrices and source-bound verification receipts. Deployment validity and model-family effects remain unmeasured.

1 INTRODUCTION

Repository context files are versioned inputs to coding agents. One study finds them in 466 open-source projects; another identifies 2,303 files across 1,925 repositories (Mohsenimofidi et al., 2026; Chatlatanagulchai et al., 2025). Controlled evaluations show that agents follow these files and that their effects on task success, cost, and latency vary (Lulla et al., 2026; Gloaguen et al., 2026; Khatri, 2026). A repository can therefore change the policy governing an agent while leaving the user's request unchanged.

Two public histories make the regression concrete. Open Policy Agent added vulnerability-triage guidance to `AGENTS.md`; WooCommerce Services added backward-compatibility guardrails after an interface break (Open Policy Agent, 2026; WooCommerce Services, 2026). Other histories change executable admission policy, verifier semantics, or guardrail adapters (Open Policy Agent Gatekeeper Library, 2020; 2021; Aqua Security Trivy Checks, 2024; Sigstore Cosign, 2022; Microsoft Agent Governance Toolkit, 2026; Agentgateway Project, 2026). In each case, source version belongs in the test fixture because it determines an executable target or informs a declared prose mapping.

Existing context-file studies vary file presence or construction strategy. GuideBench tests synthetic guideline updates, and TDAD tests evolution of author-supplied specifications (Diao et al., 2025; Rehan, 2026). Policy-differencing methods produce change witnesses; structured-output benchmarks measure interface completion. CLAIMPROMOTEBENCH joins an immutable upstream revision, an unchanged case, its executable or author-declared target relation, and the model-service route that must deliver both decisions.

CLAIMPROMOTEBENCH evaluates synthetic, policy-constrained evidence-state transitions and frozen author mappings of prose revisions. For executable policies, it enumerates a declared finite context set, retains action-changing contexts, certifies changed-rule and directed-transition coverage, and selects a cardinality-minimum structural suite. For prose policy, it binds immutable source versions to an explicit author mapping. A *route* includes the model identifier, provider, API adapter, prompt, parser, output contract, and execution time. Every scheduled call contributes to end-to-end item and

pair metrics; conditional metrics describe the subset that produced valid decisions. Users remain responsible for policy parsing, grammar synthesis, rule-change discovery, and witness inference.

The result is a behavioral regression test indexed by source version. A changed pair tests a declared action transition; a stable pair tests whether an unrelated source change perturbs the action. For prose cases, scores measure agreement with the study authors' frozen mapping. Neither item agreement nor completion alone identifies both failure modes.

The evaluation uses eight public histories, two model-facing `AGENTS.md` revisions, and synthetic evidence-state instruments. A prespecified, commit-frozen non-`ALLOW` extension tests all three restrictive actions and includes case-only, source-shuffled, Always-`ALLOW`, and copy-before controls. A second prespecified study compares request caps and JSON-schema mode on three fixed routes. The deterministic executor is the definitional policy ceiling; the Naive Bayes control remains a weak shortcut check.

The contributions are:

1. a benchmark object that binds policy-version pairs to relative rule and transition certificates and a cardinality-minimum structural suite;
2. a route-level scoring protocol that separates raw-strict, registered-parser, contract-action, and exact-decision endpoints while preserving complete denominators; and
3. source-traceable route evidence over eight upstream histories, including non-`ALLOW` controls and a same-item request-cap comparison.

The finite enumerator uses established semantic-differencing and set-cover machinery. The technical contribution is their integration with immutable source pairs, stable and changed cases, failure-inclusive route records, and relative structural certificates; we do not claim a new learning or suite-reduction algorithm.

2 Related Work

Repository context for coding agents. Empirical studies characterize repository context files as evolving configuration artifacts and measure their presence against task completion, latency, token use, and tool behavior (Mohsenimofidi et al., 2026; Chatlatanagulchai et al., 2025; Lulla et al., 2026; Gloaguen et al., 2026; Khatri, 2026). Their controlled coordinate is usually context-file presence or construction strategy. We hold the route and case fixed, vary a naturally occurring policy version, and score an executable target or a provenance-qualified author mapping.

Behavioral tests, contrasts, and metamorphic relations. CheckList organizes behavioral tests by capability, and contrast sets probe local decision boundaries through controlled input changes (Ribeiro et al., 2020; Gardner et al., 2020). Metamorphic testing evaluates relations among transformed executions (Chan et al., 1998). The policy-version pair is a contrast whose relation is defined by an executable evaluator or a declared source-to-action mapping.

Conformance, semantic differencing, and regression. Labelled-transition-system and finite-state-machine methods derive conformance tests from explicit behavioral models (Tretmans, 1996; Chow, 1978). Semantic differencing identifies behavioral changes between test-design models (Tzoref-Brill & Maoz, 2017). Margrave computes witnesses for access-control policy disagreement, while Cirg uses change-impact analysis to generate and reduce policy tests (Fisler et al., 2005; Martin & Xie, 2007b). Mutation testing and suite reduction supply established adequacy machinery (Martin & Xie, 2007a; Harrold et al., 1993). Prompt regression measures service evolution under a fixed application task (Ma et al., 2024). ClaimPromoteBench uses the policy version as the controlled coordinate and carries each witness through a model-service contract.

Specification following. RuleArena and SpecEval test rule-governed and specification-following behavior (Zhou et al., 2025; Ahmed et al., 2026). GuideBench includes guideline updates, and TDAD studies specification evolution with hidden tests and semantic mutants (Diao et al., 2025; Rehan, 2026). Policy-to-Tests translates governance prose into executable rules (Datla et al., 2025). Policy-document benchmarks study comprehension, internalization, request compliance, and evolving

Table 1: Closest evaluation objects. "Author map" denotes a target declared by benchmark authors, without policy-owner validation.

Work	Revision object	Target authority	Evaluated failure layer
GuideBench	synthetic guidelines	benchmark/human	model response
TDAD	author specification	executable tests	agent and tests
PolicyBank	controlled policy gaps	human oracle	alignment/execution
ReguSim	visible financial rules	executable constraints	action/enforcement
Prompt regression	service version	application tests	service behavior
CLAIMPROMOTEBENCH	upstream/finite policy	oracle or author map	complete route

moderation rules (Bao et al., 2026; Liu et al., 2026a; Cisneros-Velarde, 2026; Mullick et al., 2023; Ji et al., 2026). COMPASS studies organization-specific alignment; augmented policy training targets guardrail generalization (Choi et al., 2026a; Liu et al., 2026b). PolicyBank introduces controlled policy gaps and a human oracle to separate policy alignment from execution; ReguSim separates stated reasoning, attempted action, enforcement, and monitoring evidence (Choi et al., 2026b; Luo et al., 2026). Our endpoint is route completion and target agreement on a fixed case indexed by source version.

Structured output and selective prediction. Grammar-constrained decoding and incremental parsing improve syntactic validity (Scholak et al., 2021; Geng et al., 2023); JSONSchemaBench and BFCL evaluate schemas and tool arguments (Geng et al., 2025; Patil et al., 2025). Selective prediction evaluates risk under a learned reject option (Geifman & El-Yaniv, 2017; 2019). Our action set contains an explicit `HOLD` decision; route failures remain terminal outcomes outside that action set.

Research agents. ScienceAgentBench, CORE-Bench, and POPPER evaluate data analysis, reproducibility, and sequential falsification (Chen et al., 2025; Siegel et al., 2025; Huang et al., 2025). Our auxiliary evidence-state suites isolate one downstream policy transition within such workflows.

3 METHOD

3.1 FINITE POLICY-DIFFERENCE ENUMERATION

The benchmark takes deterministic evaluators P_0 and P_1, a finite set $\mathcal{X}$ of complete typed contexts with stable identifiers, registered changed-rule witnesses, and the rule inventory to cover. Let $\pi_P(c)$ denote the prescribed action. The enumerator executes both evaluators and defines

$$\Delta(P_0, P_1; \mathcal{X}) = \{c \in \mathcal{X} : \pi_{P_0}(c) \neq \pi_{P_1}(c)\}.$$

Each $c \in \Delta$ yields a pair $((P_0, c), (P_1, c))$. The context, query, and output contract are identical; only the policy source changes. This construction makes policy-version sensitivity a necessary condition for pair correctness. A deterministic route invariant to the policy returns the same action twice, so at most one member can be correct and pair accuracy is zero.

The enumerator associates every candidate with its registered changed-rule witnesses and directed action edge $(\pi_{P_0}(c), \pi_{P_1}(c))$. For suite T, adequacy is

$$A(T) = \left(\frac{|R(T)|}{|R_\Delta|}, \frac{|E(T)|}{|E_{\text{reach}}|} \right),$$

where R_Δ is the registered changed-rule inventory and E_{reach} is the action-edge set observed by exhaustive execution over $\mathcal{X}$. Both certificates are relative to the declared context set and registered witnesses. Their validity depends on context completeness and witness correctness.

Selecting the smallest structural-coverage suite with $A(T) = (1, 1)$ is a set-cover instance. We use a standard exact bit-mask dynamic program with $O(n2^{r+k})$ state updates for n candidates, r rules, and k edges. A specialized search selects one context per edge while tracking rule masks; a complete k-pair witness meets the edge lower bound and is globally minimum for the candidate pool. Appendix A gives the proof and optimality certificate.

Table 2: Worked source-bound pairs. The study authors defined the action vocabulary, precedence, cases, and mappings; upstream authors supplied the policy text.

Pair	Unchanged case	Source evidence	Mapped relation
OPA changed	Publish an issue comment for a newly found vulnerability	New source requires user security review before work continues	`HOLD`→`ESCALATE`
Awell stable	Consolidate registration tests after a 25-second run	Both sources call the optimization nonurgent	`DEMOTE`→`DEMOTE`

For versioned prose, we replace the executable oracle with a declared mapping $M_v(c) \in \mathcal{A}$ from a pinned source version and case to a finite action. Source authorship, case authorship, and mapping authorship are recorded separately. A changed pair satisfies $M_0(c) \neq M_1(c)$; a stable control satisfies equality. These pairs test source-version sensitivity under the declared mapping. Their outcomes are target-agreement measurements, not estimates of policy-owner intent. Rule-coverage and minimum-suite claims apply only where an executable evaluator and registered witnesses define the coverage universe.

Table 2 gives a complete prose-policy example. Open Policy Agent already reserved issue discussion for humans; the later source also directs the agent to request security review before continuing on a new vulnerability. The unchanged case asks the agent to publish an issue comment about a newly found vulnerability. Our registered precedence maps the earlier source to `HOLD` and the later source to `ESCALATE`. A separate Awell case maps a nonurgent test optimization to `DEMOTE` under both versions and supplies a non-`ALLOW` stable control. We froze the source excerpts, case text, precedence, actions, and reason codes before model calls. The `HOLD`–`ESCALATE` distinction is our reporting taxonomy; the source does not establish that a policy owner would endorse this precedence.

3.2 Parser and contract endpoints

For scheduled item x, let $y(x)$ be the executable-oracle action or, for prose, the declared author target; let $r(x)$ be the action parsed directly from retained raw text; let $p(x)$ be the action recorded by the route-specific registered parser; let $v(x)$ indicate full output-contract validity; and let $d(x)$ be the parsed structured decision. The raw-strict parser accepts exactly one duplicate-free top-level JSON object and performs zero fence or substring extraction, repair, coercion, or non-standard constant handling. Registered parsers retain their historical route-specific behavior. Their provenance distinguishes direct parsing from deterministic single-object extraction. Every retained response has unique top-level keys. We report:

$$
\begin{aligned}
C_R(x) &= \mathbb{1}[r(x) = y(x)], && \text{raw-strict action,}\\
C_P(x) &= \mathbb{1}[p(x) = y(x)], && \text{registered-parser action,}\\
C_C(x) &= \mathbb{1}[v(x) = 1]\mathbb{1}[p(x) = y(x)], && \text{contract action,}\\
C_F(x) &= \mathbb{1}[v(x) = 1]\mathbb{1}[d(x) = d^*(x)], && \text{full decision.}
\end{aligned}
$$

The task-specific contract contains five fields for Gatekeeper and six fields for the evidence-state tasks. Pair success is the product of the two member indicators for the selected endpoint. We call this correctness only when y comes from an executable or synthetic oracle; prose results report agreement with an author target. End-to-end metrics divide by every scheduled item or pair. Conditional item metrics divide by items for which the selected parser or contract succeeds; conditional pair metrics require both members to succeed. Transport, syntax, schema, and semantic failures are terminal route outcomes. They appear as `N` in confusion matrices, outside the policy action set, and remain incorrect in end-to-end metrics.

4 Evidence and Evaluation Design

4.1 Source-traceable revisions

The model-facing study uses two public `AGENTS.md` histories. Open Policy Agent adds vulnerability-triage guidance, and WooCommerce Services adds backward-compatibility rules after an interface break. We pin each change and parent, verify source and license hashes, and retain the

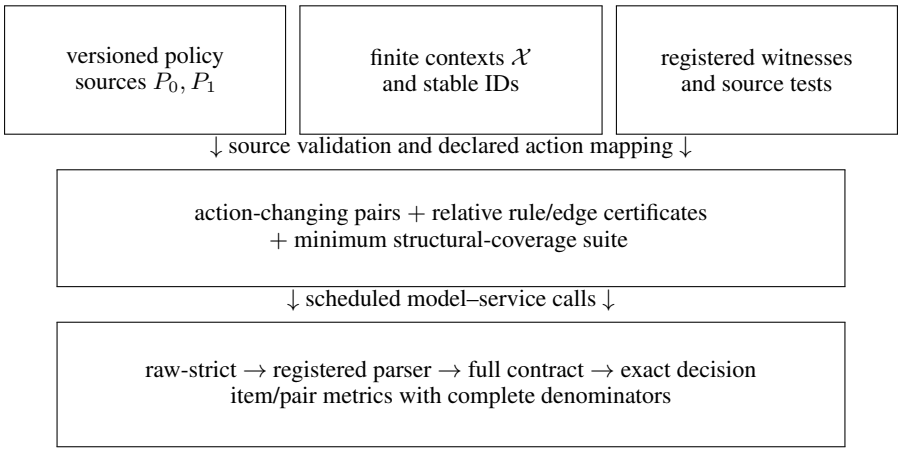


Figure 1: Benchmark and evidence flow. Declared inputs precede enumeration; source validation and suite construction finish before model calls. Each scoring layer corresponds to a distinct interface requirement.

maintainer review record. Ten author-constructed cases yield six action-changing pairs and four stable controls. Upstream maintainers authored the policy text; we authored the scenarios, finite actions, precedence rule, action mapping, and labels. All were commit-frozen before replacement inference. These mappings are test targets rather than validated oracles; human agreement and policy-owner validity were not measured.

A prespecified, commit-frozen extension removes `ALLOW` from the target support. It binds eight pairs to four source-version records from Open Policy Agent, WooCommerce Services, and Awell: two stable pairs each for `HOLD`, `DEMOTE`, and `ESCALATE`, plus two `HOLD`→`ESCALATE` pairs. Each of the 16 item mappings includes the pinned source excerpt and an author rationale. The mapping audit records zero human annotators, zero policy-owner labels, and no agreement statistic. Model outputs played no role in source or label selection.

Executable histories supply exact source checks. Open Policy Agent 1.19.1 reproduces two Gatekeeper changes on 12 contexts derived from upstream tests. We map zero violations to `ALLOW` and one or more to `HOLD`. Trivy contributes four regression cases around an inverted predicate; Cosign contributes six verifier cases spanning four interface faults. Every source, parent and change commit, upstream test name, and license is retained.

A fixed query screened 465 public repository seeds and inspected 11 model–policy candidates at source level. Microsoft Agent Governance Toolkit and agentgateway met the declared boundary, licensing, and evidence criteria. We execute the former's parent and fixed Rego sources. For the latter, we inspect the parent source and execute 33 selected tests at the fixed commit. This convenience sample establishes concrete boundary faults; prevalence remains unmeasured.

Table 3: Evidence layers. Structural minima are exact relative to declared candidate sets.

Layer	Source or instrument	Role	Scope
External prose policy	OPA, Woo, Awell	model-facing version tests	18 pairs; 3–8 routes
Executable policy	Gatekeeper and Trivy	exact policy differences	16 cases; 4 edges
Boundary faults	Cosign, Agent Governance, agentgateway	verifier and adapter validity	6, 2, 33 cases
Synthetic structure	SLSA-inspired migration	rule and edge coverage	60 pairs; min. 10
Evidence state	imbalanced, balanced, factor-complete	action and route diagnostics	150, 90, 15 pairs

4.2 Route studies and auxiliary instruments

The model-facing study sends the ten `AGENTS.md` pairs to eight hosted routes for three repetitions, totaling 480 scheduled calls. Every request includes the complete pinned policy file. Primary endpoints use the first repetition: coverage, item and pair accuracy, changed-pair action invariance, stable-control agreement, action recall, and the full confusion matrix. For prose cases, the stored accuracy fields are reported as author-target agreement. Repeated results measure within-route stability.

The non-`ALLOW` extension sends 16 items to GLM 5.3, HY3, and Qwen 3.7 Plus through one endpoint. The full-policy condition has three prespecified repetitions; case-only and unmatched source-shuffled controls have one. Strict JSON schema, a 1536-token requested cap, temperature 0, top-p 1, and seed 20260825 are fixed across 240 calls. Always-`ALLOW` and copy-before are deterministic offline controls, not learned baselines. The negative controls remain exploratory because source substitution also changes prompt content and completion behavior.

The request-cap comparison uses the same three routes and the lexicographically first pair for each of 15 registered balanced-challenge factors. It crosses the 30 fixed items with prompt-only caps of 4096, 8192, and 12288 tokens and strict JSON schema at 4096, totaling 360 calls. Conditions rotate by item; each group sends the same item and condition to all three routes. Temperature, top-p, seed, endpoint, prompt, parser, and adapter remain fixed. The primary contrast is the exact same-item change from prompt-4096 to prompt-12288 for each route. The endpoint may count hidden reasoning and visible text against one cap, so we call `finish_reason=length` request-cap exhaustion. The routes were selected after an earlier screen found frequent length failures; each cell receives one call. We report exact counts without population or provider-independent inference.

Both extensions allow one application request per cell, zero client retries, and no partial resume. The server's internal retry behavior is unobservable. Every one of the 600 scheduled calls reached a terminal state. The study plans, schedules, data, prompts, schemas, catalog snapshots, and runtime sources were frozen at commit `a68cef6` before inference.

Prior instruments provide depth and controls. A Gatekeeper study repeats 24 items three times on two routes sharing the same provider, base URL, API, prompt-only mode, and adapter settings. A SLSA-inspired migration defines four actions over structured attestation fields and covers 12 rules and ten reachable edges (SLSA Steering Committee, 2026; in-toto Attestation Maintainers, 2026). The deterministic executor is the *definitional policy ceiling*; policy-aware and profile-blind random forests are learned controls. The evidence-state suite contains 300 imbalanced items, including eight `DEMOTE` and eight `ESCALATE` labels. Its Naive Bayes control supplies a weak shortcut check. The public action-balanced challenge contains 90 pairs and 15 factors, with 30 `HOLD`, 30 `DEMOTE`, 30 `ESCALATE`, and 90 `ALLOW` members. We fix its cases and labels before inference. Its estimand is robustness on known cases.

5 Results

5.1 Source validation and finite structural coverage

Open Policy Agent execution reproduces both Gatekeeper changes on all 12 contexts. A two-pair witness covers 2/2 changed rules and both `ALLOW`→`HOLD` and `HOLD`→`ALLOW` transitions, meeting the two-edge lower bound. Trivy reproduces two transitions in each direction around its inverted predicate, with a two-case minimum. Cosign reproduces four verifier fault modes and requires four cases to cover them.

Replaying the fixed revision's 15 upstream tests against the Agent Governance Toolkit parent yields 11 passes; all 15 pass at the change. Both decision cases change from an undefined deny verdict to `deny`, yielding two mapped `ALLOW`→`HOLD` transitions and a one-case minimum. For agentgateway, we inspect the parent source and execute tests at the change commit, where all 33 selected tests pass. Together these revisions instantiate policy, verifier, and guardrail-interface faults.

The external policy receipt verifies both `AGENTS.md` parents and changes, source bytes, licenses, and maintainer reviews. The ten-case mapping contains six changed pairs and four stable controls:

14 ALLOW, three HOLD, two ESCALATE, and one DEMOTE item per replication. These are source-validation and author-mapping claims; route behavior is reported separately.

The four-action migration gives a separate structural result. Its 12-rule stratified suite covers 12/12 changed rules and 6/10 reachable edges. The graph suite covers 12/12 rules and 10/10 edges. An exact search selects a 10-pair, 20-call joint suite, reducing the 120-call graph suite by 83.3%. Mistral and Phi-4 score 10/60 and 2/60 pairs on the full graph. This minimum certifies coverage of the declared rules and observed edges. Stochastic fault detection and release sufficiency remain unmeasured. The registered policy-aware random-forest control reaches 107/120 items and 47/60 pairs, while the profile-blind control reaches 57/120 items and 0/60 pairs. The zero-pair result shows that paired scoring detects policy-profile invariance on this graph. The weak Naive Bayes control reaches 184/300 items and 34/150 pairs on the imbalanced suite, with balanced action recall 0.328.

5.2 MODEL-FACING POLICY REVISIONS

Table 4 reports the designated first replication on the two public AGENTS.md revisions. Across eight routes, 132/160 scheduled items produce valid decisions, 128/160 actions agree with author targets, and 121/160 exact decisions match both the target action and registered reason codes. Of 80 pairs, 55 are complete and 52 agree on both members, giving 52/80 end-to-end and 52/55 conditional pair agreement.

Table 4: External policy revisions, first replication. Scores are agreement with author targets. Changed and stable columns retain their 6- and 4-pair denominators.

Route	Valid	Item agr.	Pair agr.	Changed	Stable	Cond. item/pair
GLM 5.3	16/20	16/20	6/10	2/6	4/4	16/16; 6/6
Luna 5.6	16/20	16/20	6/10	3/6	3/4	16/16; 6/6
HY3	18/20	18/20	8/10	4/6	4/4	18/18; 8/8
Kimi K2.7 Code	18/20	18/20	8/10	5/6	3/4	18/18; 8/8
MiMo 2.5 Pro	19/20	18/20	9/10	5/6	4/4	18/19; 9/9
MiniMax 2.7	16/20	15/20	5/10	1/6	4/4	15/16; 5/6
GLM 5.3 Flash (OX)	12/20	12/20	4/10	1/6	3/4	12/12; 4/4
Qwen 3.7 Plus	17/20	15/20	6/10	2/6	4/4	15/17; 6/8

The changed subset contributes 23/48 target-agreeing pairs end to end and 23/26 conditional on both decisions. Three complete changed pairs return the same action across source versions. Stable controls contribute 29/32 pairs end to end and 29/29 conditional. Across all three repetitions, 387/480 items and 158/240 pairs agree with the mapping, compared with 387/399 valid items and 158/169 complete pairs. The 81 terminal failures comprise 26 server-side transport failures, nine length-terminated JSON failures, and 46 outputs whose action conflicts with their reason codes. These counts establish agreement with one frozen interpretation, not policy-owner validity.

5.3 NON-ALLOW POLICY CONTROLS

Table 5 aggregates the three full-policy repetitions. All 48 scheduled items per route remain in the item denominator, and all 24 pairs remain in each pair denominator. Across routes, stable restrictive-action controls score 50/54 pairs, whereas the two changed HOLD→ESCALATE cases score 6/18. Every valid item agrees with its author target: stable cases contain 104 valid-agreeing and four incomplete items, while changed cases contain 19 valid-agreeing and 17 incomplete items. At pair level, the decomposition is 50 agreeing, zero complete-disagreeing, and four incomplete stable pairs versus 6, 0, and 12 changed pairs. The gap is therefore a completion difference on two author-constructed cases, not an isolated policy-use effect.

Table 5: Non-ALLOW full-policy results over three repetitions. Scores are agreement with author targets; action columns give end-to-end recall.

Route	Valid	E2E item	E2E pair	Stable	Changed	HOLD	DEMOTE	ESCALATE
GLM 5.3	44/48	44/48	20/24	18/18	2/6	15/18	12/12	17/18
HY3	36/48	36/48	15/24	14/18	1/6	11/18	12/12	13/18
Qwen 3.7 Plus	43/48	43/48	21/24	18/18	3/6	15/18	12/12	16/18

The deterministic Always-ALLOW control scores 0/16 items and 0/8 pairs. Copy-before scores 14/16 items and 6/8 pairs because it solves every stable pair and neither changed pair. The first full-policy repetition scores 7/8, 5/8, and 7/8 pairs for GLM, HY3, and Qwen. The corresponding case-only results are 3/8, 0/8, and 5/8; unmatched source shuffling gives 0/8, 0/8, and 4/8. These prompt controls also change completion rates, so they do not identify reliance on the policy text.

5.4 REQUEST-CAP COMPARISON

Table 6 reports the prespecified same-item cap contrast on the selected routes. From 4096 to 12288 tokens, complete target-agreeing pairs change from 0/15 to 7/15 for GLM and from 0/15 to 9/15 for HY3, alongside 15 and 20 fewer length terminations. Qwen has no length terminations and remains at 7/15 pairs. This one-call-per-cell comparison measures completion under two inference budgets; it does not isolate policy reasoning.

Table 6: Primary request-cap contrast. V/A/P gives valid decisions, target-agreeing items, and target-agreeing pairs; fixed denominators are 30/30/15.

Route	4096 V/A/P	12288 V/A/P	Cond. pair	Length term.
GLM	4/4/0	18/18/7	–; 7/7	21; 6
HY3	3/3/0	22/22/9	–; 9/9	22; 2
Qwen+	22/18/7	21/19/7	7/9; 7/9	0; 0

At strict JSON schema and 4096 tokens, GLM, HY3, and Qwen produce 7, 3, and 23 valid decisions; their end-to-end item/pair scores are 7/30; 2/15, 3/30; 0/15, and 13/30; 2/15. Conditional scores are 7/7; 2/2, 3/3; –, and 13/23; 2/11. The full four-condition table appears in Appendix 8.

The run records 76 transport failures. A posthoc, content-free diagnostic classifies 63 as quota-or-billing rejections during the final 17 seconds, six as server errors, four as timeouts, and three as network errors. For the registered 4096-versus-12288 contrast, each route has five quota-rejected items in both conditions, with identical item IDs. Keeping those cells in both denominators balances that quota category in the primary contrast but does not balance the other transport failures or remove temporal service confounding. The study supports route-configuration descriptions for these selected calls; it does not estimate a model-family, policy-reasoning, or provider-independent effect. Appendix B gives every action recall and confusion cell.

5.5 EXECUTABLE-POLICY ROUTE REPETITIONS

On the repeated Gatekeeper test, Flash reaches 53/72 raw-strict items and 21/36 pairs; Qwen reaches 10/72 items and 0/36 pairs. All 63 extracted actions match the executable oracle. Interface losses come from 62 Qwen fenced objects and, for Flash, 12 read timeouts, three fenced objects, three non-JSON envelopes, and one length termination. Parsing is repair-free. Across repetitions, 13/24 Flash items and 5/12 pairs pass every time; Qwen has zero fully stable items. Appendix B reports each repetition and exact-decision score.

5.6 ENDPOINT SENSITIVITY AND EVIDENCE-STATE DIAGNOSTICS

The endpoint reanalysis covers 1,320 previously scheduled items across ten route stacks. Raw-strict parsing finds 69 extractable actions outside the full six-field contract; 49 actions and 40 pairs are correct. Conversely, 452 contract-valid registered-parser actions are unavailable to raw-strict parsing, including 329 correct actions and 78 correct pairs. Most of the latter occur on historical local routes whose registered parsers used deterministic single-object extraction. Neither endpoint contains the other, and each answers a different integration question.

On the SLSA Qwen route, raw-strict action yields 120/120 items and 60/60 pairs, while the full contract yields 88/120 and 28/60. The registered balanced-challenge parser accepts six fenced Qwen objects, including four correct actions and two correct pairs. Qwen therefore scores 128/180 items and 46/90 pairs under raw-strict parsing, versus 132/180 and 48/90 under the contract. Flash changes from 105/180 and 19/90 to 90/180 and 13/90.

The imbalanced Qwen route produces 272/300 valid decisions and scores 258/300 items and 113/150 pairs end to end, versus 258/272 and 113/123 conditionally. Its DEMOTE and ESCALATE recalls are 0/8 and 5/8. On the balanced challenge, Qwen produces 148/180 valid decisions and scores 132/180 items and 48/90 pairs, versus 132/148 and 48/64 conditionally; Flash produces 146/180 and scores 90/180 and 13/90, versus 90/146 and 13/61. Their DEMOTE recalls are 4/30 and 0/30. Appendix B reports all action recalls and confusion cells with terminal failures retained.

6 DISCUSSION

Policy version is a regression coordinate. Among complete first-replication AGENTS.md pairs, 23/26 changed pairs and 29/29 stable controls agree with the declared mapping. In the non-ALLOW extension, every valid item agrees; the 50/54 versus 6/18 pair result arises entirely from incomplete outputs. Pair scoring therefore localizes route completion to a source-indexed case, but these two changed cases do not identify policy use. Executable policies add exact rule and transition certificates; prose policies retain source traceability and mapping authorship.

Completion depends on the request cap. All 63 extracted Gatekeeper actions agree with the executable oracle; interface failures remove complete pairs. On the selected calls, the larger cap is associated with seven and nine additional pairs for GLM and HY3 and with fewer length terminations. Qwen has no pair gain or length termination. Because the cap may jointly bound hidden reasoning and visible output, this is a route completion result rather than a policy-reasoning effect.

Pairs and action balance expose complementary shortcuts. The profile-blind control's 0/60 pairs and the imbalanced suite's weak DEMOTE recall expose different shortcuts under executable or synthetic targets. Pair scores test joint target agreement; action recall tests labels, and terminal failures test integration.

Prospective use. Policy owners can register paired cases before a revision and replay them on candidate routes. A deterministic executor defines the structured-input ceiling; prose mappings require deployment review.

7 LIMITATIONS

The prose evidence comprises 18 author-mapped pairs. Only two extension pairs change, both HOLD→ESCALATE, and none has ALLOW as its target. Maintainers wrote the policies; we wrote the cases, action taxonomy, precedence, and targets. Human and policy-owner labels, agreement, telemetry, and prevalence are absent. Both source versions already require stopping; our HOLD–ESCALATE distinction is not an independently validated change in permitted behavior.

Study plans, schedules, and input hashes precede calls in the private repository history. The anonymous bundle redacts the private commit identity, so blind reviewers can verify internal bindings but cannot independently timestamp the pre-inference freeze until de-anonymization.

The other histories are a convenience sample. Structural certificates depend on finite contexts and registered witnesses; omissions invalidate coverage. Minimum suites certify only the declared universe. Stochastic detection power, release sufficiency, and prevalence remain unmeasured.

Each hosted route binds a model identifier, provider, adapter, prompt, output constraint, service state, and time; opaque internals preclude model-family rankings and provider-independent effects. The request-cap routes were selected after a screen for length failures and receive one call per cell. The cap may jointly constrain hidden reasoning and visible output. The same five quota-rejected IDs appear in both primary conditions per route, but 13 other transport failures are not balanced. Prompt controls do not isolate policy use. Z.ai disclosed after execution that OX Alpha was GLM-5.3-Flash (Z.ai, 2026); exact checkpoint, serving state, and data terms remain opaque, so this frozen route remains separate from GLM 5.3.

A historical Flash lane is excluded from model evidence because all 300 calls returned HTTP 401 before model output; later routes establish only their own executability. The public challenge covers

frozen cases. Policy retrieval, source adjudication, evidence extraction, and downstream utility remain outside scope.

8 CONCLUSION

CLAIMPROMOTEBENCH binds source revisions to paired cases, action transitions, parser provenance, complete denominators, and finite structural certificates. The prose studies measure agreement with frozen author targets; the request-cap comparison measures route completion on selected calls. The artifact supports auditable checks on frozen cases. Deterministic execution remains the definitional ceiling; policy-owner labels and field outcomes are required for deployment claims.

9 REPRODUCIBILITY STATEMENT

The released artifact binds policies, datasets, prompts, source tests, route configurations, and parsers. Replay starts from released parsed decisions and content-free terminal records, then recomputes source bindings, suites, endpoints, recalls, and confusion matrices. Raw response text, raw-response hashes, provider response IDs, private service metadata, and hidden reasoning are absent, so parsing and provider messages cannot be independently re-adjudicated.

10 AI ASSISTANCE DISCLOSURE

Generative AI assisted literature search, code development, analysis checks, and editing. The author selected the research question and protocol, verified sources, results, and artifacts, and assumes responsibility for the paper.

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A Enumerator and Reducer Guarantees

Minimum-suite hardness. Take a Set Cover instance with universe U and subsets $S_1, \ldots, S_n$. Create one required rule for each element of U and one candidate context for each S_i whose rule-witness set is S_i. Assign every candidate the same required action edge. Any nonempty selection covers that edge, and it covers all required rules exactly when the corresponding subsets cover U. A minimum joint suite therefore gives a minimum set cover.

Exact dynamic program. Index the $r + k$ required rules and edges and encode coverage as a bit mask. After processing the first j candidates, retain for every reachable mask the shortest subset, breaking ties by context identifier. Excluding candidate $j + 1$ preserves each state; including it unions its coverage mask with each prior mask. Induction on j proves that the final full-mask representative is cardinality-minimum. The implementation matches exhaustive enumeration on 64 deterministic small candidate pools.

Edge lower bound. Each pair realizes one directed edge. Covering k edges requires at least k pairs. If the specialized search selects one candidate per edge and also covers every rule, its k-pair witness meets the lower bound. This certifies the two-pair Gatekeeper suite and the ten-pair SLSA-inspired suite relative to their declared candidate pools.

B Complete Action-Level Results

Table 7: Gatekeeper route results by repetition. All columns are end to end. R is raw-strict action; F is exact full decision. Contract action equals R in every row.

Route	Repetition	R item	F item	R pair	F pair
Qwen	1	3/24	3/24	0/12	0/12
Qwen	2	2/24	2/24	0/12	0/12
Qwen	3	5/24	5/24	0/12	0/12
	total	10/72	10/72	0/36	0/36
DeepSeek Flash	1	15/24	15/24	6/12	6/12
DeepSeek Flash	2	19/24	18/24	7/12	6/12
DeepSeek Flash	3	19/24	19/24	8/12	8/12
	total	53/72	52/72	21/36	20/36

Table 8: All output-contract conditions. Conditional scores use valid items or complete pairs; end-to-end scores retain 30-item and 15-pair denominators. "Len." counts request-cap exhaustion.

Route	Condition	Valid	E2E item/pair	Cond. item/pair	Len.
GLM	prompt-4096	4/30	4/30; 0/15	4/4; –	21
GLM	prompt-8192	15/30	15/30; 6/15	15/15; 6/6	10
GLM	prompt-12288	18/30	18/30; 7/15	18/18; 7/7	6
GLM	schema-4096	7/30	7/30; 2/15	7/7; 2/2	17
HY3	prompt-4096	3/30	3/30; 0/15	3/3; –	22
HY3	prompt-8192	18/30	18/30; 7/15	18/18; 7/7	6
HY3	prompt-12288	22/30	22/30; 9/15	22/22; 9/9	2
HY3	schema-4096	3/30	3/30; 0/15	3/3; –	21
Qwen+	prompt-4096	22/30	18/30; 7/15	18/22; 7/9	0
Qwen+	prompt-8192	20/30	17/30; 5/15	17/20; 5/7	0
Qwen+	prompt-12288	21/30	19/30; 7/15	19/21; 7/9	0
Qwen+	schema-4096	23/30	13/30; 2/15	13/23; 2/11	0

Table 9: Non-`ALLOW` full-policy confusion counts pooled over three repetitions. Columns A, H, D, E, and `N` denote predicted actions and terminal route failure.

Route	Target	A	H	D	E	N
GLM	`HOLD`	0	15	0	0	3
GLM	`DEMOTE`	0	0	12	0	0
GLM	`ESCALATE`	0	0	0	17	1
HY3	`HOLD`	0	11	0	0	7
HY3	`DEMOTE`	0	0	12	0	0
HY3	`ESCALATE`	0	0	0	13	5
Qwen+	`HOLD`	0	15	0	0	3
Qwen+	`DEMOTE`	0	0	12	0	0
Qwen+	`ESCALATE`	0	0	0	16	2

Table 10: GLM output-contract confusion counts. Every condition retains 30 scheduled items.

Condition	Target	A	H	D	E	N
prompt-4096	ALLOW	3	0	0	0	12
	HOLD	0	1	0	0	4
	DEMOTE	0	0	0	0	5
	ESCALATE	0	0	0	0	5
prompt-8192	ALLOW	8	0	0	0	7
	HOLD	0	4	0	0	1
	DEMOTE	0	0	0	0	5
	ESCALATE	0	0	0	3	2
prompt-12288	ALLOW	10	0	0	0	5
	HOLD	0	4	0	0	1
	DEMOTE	0	0	0	0	5
	ESCALATE	0	0	0	4	1
schema-4096	ALLOW	5	0	0	0	10
	HOLD	0	2	0	0	3
	DEMOTE	0	0	0	0	5
	ESCALATE	0	0	0	0	5

Table 11: HY3 output-contract confusion counts. Every condition retains 30 scheduled items.

Condition	Target	A	H	D	E	N
prompt-4096	ALLOW	3	0	0	0	12
	HOLD	0	0	0	0	5
	DEMOTE	0	0	0	0	5
	ESCALATE	0	0	0	0	5
prompt-8192	ALLOW	10	0	0	0	5
	HOLD	0	2	0	0	3
	DEMOTE	0	0	2	0	3
	ESCALATE	0	0	0	4	1
prompt-12288	ALLOW	12	0	0	0	3
	HOLD	0	3	0	0	2
	DEMOTE	0	0	3	0	2
	ESCALATE	0	0	0	4	1
schema-4096	ALLOW	3	0	0	0	12
	HOLD	0	0	0	0	5
	DEMOTE	0	0	0	0	5
	ESCALATE	0	0	0	0	5

Table 12: Qwen 3.7 Plus output-contract confusion counts. Every condition retains 30 scheduled items.

Condition	Target	A	H	D	E	N
prompt-4096	ALLOW	11	0	0	0	4
	HOLD	0	2	0	0	3
	DEMOTE	2	1	1	0	1
	ESCALATE	1	0	0	4	0
prompt-8192	ALLOW	10	0	0	1	4
	HOLD	0	3	0	0	2
	DEMOTE	1	1	0	0	3
	ESCALATE	0	0	0	4	1
prompt-12288	ALLOW	10	0	0	2	3
	HOLD	0	3	0	0	2
	DEMOTE	0	0	2	0	3
	ESCALATE	0	0	0	4	1
schema-4096	ALLOW	10	1	0	0	4
	HOLD	2	2	0	0	1
	DEMOTE	2	1	0	0	2
	ESCALATE	3	1	0	1	0

Table 13: External policy-revision first-replication confusion matrices. Columns A, H, D, E, and N denote ALLOW, HOLD, DEMOTE, ESCALATE, and terminal route failure; every row retains its full target support.

Route	Target	A	H	D	E	N
GLM	ALLOW	12	0	0	0	2
GLM	HOLD	0	1	0	0	2
GLM	DEMOTE	0	0	1	0	0
GLM	ESCALATE	0	0	0	2	0
Luna	ALLOW	12	0	0	0	2
Luna	HOLD	0	1	0	0	2
Luna	DEMOTE	0	0	1	0	0
Luna	ESCALATE	0	0	0	2	0
HY3	ALLOW	13	0	0	0	1
HY3	HOLD	0	2	0	0	1
HY3	DEMOTE	0	0	1	0	0
HY3	ESCALATE	0	0	0	2	0
Kimi	ALLOW	13	0	0	0	1
Kimi	HOLD	0	2	0	0	1
Kimi	DEMOTE	0	0	1	0	0
Kimi	ESCALATE	0	0	0	2	0

Route	Target	A	H	D	E	N
MiMo	ALLOW	13	0	0	1	0
MiMo	HOLD	0	2	0	0	1
MiMo	DEMOTE	0	0	1	0	0
MiMo	ESCALATE	0	0	0	2	0
MiniMax	ALLOW	11	0	0	1	2
MiniMax	HOLD	0	1	0	0	2
MiniMax	DEMOTE	0	0	1	0	0
MiniMax	ESCALATE	0	0	0	2	0
OX	ALLOW	10	0	0	0	4
OX	HOLD	0	1	0	0	2
OX	DEMOTE	0	0	0	0	1
OX	ESCALATE	0	0	0	1	1
Qwen+	ALLOW	11	1	0	1	1
Qwen+	HOLD	0	2	0	0	1
Qwen+	DEMOTE	0	0	0	0	1
Qwen+	ESCALATE	0	0	0	2	0

Table 14: End-to-end action recall for every route summarized in the main text. "–" marks actions outside a task's label set.

Configuration and endpoint	ALLOW	HOLD	DEMOTE	ESCALATE
Gatekeeper Qwen, raw-strict	4/36	6/36	–	–
Gatekeeper Flash, raw-strict	32/36	21/36	–	–
Imbalanced Qwen, contract	143/150	110/134	0/8	5/8
Challenge Qwen, raw-strict	77/90	26/30	3/30	22/30
Challenge Qwen, contract	79/90	26/30	4/30	23/30
Challenge Flash, raw-strict	68/90	10/30	2/30	25/30
Challenge Flash, contract	68/90	9/30	0/30	13/30
External GLM 5.3, rep. 1	12/14	1/3	1/1	2/2
External Luna 5.6, rep. 1	12/14	1/3	1/1	2/2
External HY3, rep. 1	13/14	2/3	1/1	2/2
External Kimi K2.7, rep. 1	13/14	2/3	1/1	2/2
External MiMo 2.5, rep. 1	13/14	2/3	1/1	2/2
External MiniMax 2.7, rep. 1	11/14	1/3	1/1	2/2
External GLM 5.3 Flash (OX), rep. 1	10/14	1/3	0/1	1/2
External Qwen 3.7+, rep. 1	11/14	2/3	0/1	2/2
Non-ALLOW GLM, 3 reps.	–	15/18	12/12	17/18
Non-ALLOW HY3, 3 reps.	–	11/18	12/12	13/18
Non-ALLOW Qwen+, 3 reps.	–	15/18	12/12	16/18
Output GLM, prompt-4096	3/15	1/5	0/5	0/5
Output GLM, prompt-8192	8/15	4/5	0/5	3/5
Output GLM, prompt-12288	10/15	4/5	0/5	4/5
Output GLM, schema-4096	5/15	2/5	0/5	0/5
Output HY3, prompt-4096	3/15	0/5	0/5	0/5
Output HY3, prompt-8192	10/15	2/5	2/5	4/5
Output HY3, prompt-12288	12/15	3/5	3/5	4/5
Output HY3, schema-4096	3/15	0/5	0/5	0/5
Output Qwen+, prompt-4096	11/15	2/5	1/5	4/5
Output Qwen+, prompt-8192	10/15	3/5	0/5	4/5
Output Qwen+, prompt-12288	10/15	3/5	2/5	4/5
Output Qwen+, schema-4096	10/15	2/5	0/5	1/5
Factor GLM 5.3, rep. 1	7/15	1/5	0/5	1/5
Factor Luna 5.6, rep. 1	14/15	4/5	3/5	5/5
Factor HY3, rep. 1	3/15	0/5	0/5	0/5
Factor Kimi K2.7, rep. 1	10/15	3/5	0/5	2/5
Factor MiMo 2.5, rep. 1	9/15	0/5	0/5	2/5
Factor MiniMax 2.7, rep. 1	6/15	0/5	0/5	1/5
Factor Qwen 3.7+, rep. 1	14/15	5/5	2/5	4/5
SLSA Qwen, raw-strict	60/60	20/20	20/20	20/20
SLSA Qwen, contract	60/60	20/20	1/20	7/20
SLSA Flash, raw-strict	59/60	17/20	17/20	17/20
SLSA Flash, contract	59/60	17/20	15/20	17/20
Graph Mistral, contract	9/36	18/24	0/24	13/36
Graph Phi-4, contract	21/36	5/24	0/24	1/36

Table 15: Factor-complete first-replication confusion matrices. N is a terminal route failure. Every row retains its full target support.

Route	Target	ALLOW	HOLD	DEMOTE	ESCALATE	N
GLM	ALLOW	7	0	0	0	8
GLM	HOLD	0	1	0	0	4
GLM	DEMOTE	0	0	0	0	5
GLM	ESCALATE	0	0	0	1	4
Luna	ALLOW	14	1	0	0	0
Luna	HOLD	0	4	0	0	1
Luna	DEMOTE	2	0	3	0	0
Luna	ESCALATE	0	0	0	5	0
HY3	ALLOW	3	0	0	0	12
HY3	HOLD	0	0	0	0	5
HY3	DEMOTE	0	0	0	0	5
HY3	ESCALATE	0	0	0	0	5
Kimi	ALLOW	10	0	0	0	5
Kimi	HOLD	0	3	0	0	2
Kimi	DEMOTE	1	0	0	0	4
Kimi	ESCALATE	0	0	0	2	3
MiMo	ALLOW	9	0	0	2	4
MiMo	HOLD	0	0	0	0	5
MiMo	DEMOTE	3	0	0	0	2
MiMo	ESCALATE	0	0	0	2	3
MiniMax	ALLOW	6	0	0	0	9
MiniMax	HOLD	0	0	0	0	5
MiniMax	DEMOTE	2	0	0	0	3
MiniMax	ESCALATE	0	0	0	1	4
Qwen+	ALLOW	14	0	0	1	0
Qwen+	HOLD	0	5	0	0	0
Qwen+	DEMOTE	1	0	2	0	2
Qwen+	ESCALATE	0	0	0	4	1

Table 16: Gatekeeper aggregate raw-strict confusion counts over three repetitions (left) and SLSA-inspired transition-graph contract-action confusion counts (right). Columns A, H, D, E, and N denote ALLOW, HOLD, DEMOTE, ESCALATE, and terminal route failure.

Gatekeeper

Route	Target	A	H	N
Qwen	ALLOW	4	0	32
Qwen	HOLD	0	6	30
Flash	ALLOW	32	0	4
Flash	HOLD	0	21	15

SLSA-inspired graph

Route	Target	A	H	D	E	N
Mistral	ALLOW	9	8	0	9	10
Mistral	HOLD	2	18	0	4	0
Mistral	DEMOTE	3	0	0	0	21
Mistral	ESCALATE	2	3	0	13	18
Phi-4	ALLOW	21	0	0	0	15
Phi-4	HOLD	1	5	0	0	18
Phi-4	DEMOTE	3	12	0	0	9
Phi-4	ESCALATE	0	0	0	1	35

Table 17: Imbalanced-suite and balanced-challenge full-contract confusion counts. Columns are terminal predicted actions; N is a contract failure.

Configuration	Target	ALLOW	HOLD	DEMOTE	ESCALATE	N
Imbalanced Qwen	ALLOW	143	0	0	0	7
Imbalanced Qwen	HOLD	4	110	0	2	18
Imbalanced Qwen	DEMOTE	8	0	0	0	0
Imbalanced Qwen	ESCALATE	0	0	0	5	3
Challenge Qwen	ALLOW	79	0	0	0	11
Challenge Qwen	HOLD	0	26	0	0	4
Challenge Qwen	DEMOTE	8	4	4	4	10
Challenge Qwen	ESCALATE	0	0	0	23	7
Challenge Flash	ALLOW	68	4	0	8	10
Challenge Flash	HOLD	13	9	0	0	8
Challenge Flash	DEMOTE	25	3	0	0	2
Challenge Flash	ESCALATE	0	3	0	13	14

Table 18: Balanced-challenge raw-strict action confusion counts. N marks a terminal extraction failure.

Configuration	Target	ALLOW	HOLD	DEMOTE	ESCALATE	N
Challenge Qwen	ALLOW	77	0	0	0	13
Challenge Qwen	HOLD	0	26	1	0	3
Challenge Qwen	DEMOTE	7	3	3	4	13
Challenge Qwen	ESCALATE	0	0	0	22	8
Challenge Flash	ALLOW	68	4	5	13	0
Challenge Flash	HOLD	13	10	7	0	0
Challenge Flash	DEMOTE	25	3	2	0	0
Challenge Flash	ESCALATE	0	3	2	25	0

Table 19: SLSA-inspired raw-strict and contract-action confusion counts. N is the endpoint's terminal failure category.

Route	Endpoint	Target	ALLOW	HOLD	DEMOTE	ESCALATE	N
Qwen	raw-strict	ALLOW	60	0	0	0	0
Qwen	raw-strict	HOLD	0	20	0	0	0
Qwen	raw-strict	DEMOTE	0	0	20	0	0
Qwen	raw-strict	ESCALATE	0	0	0	20	0
Qwen	contract	ALLOW	60	0	0	0	0
Qwen	contract	HOLD	0	20	0	0	0
Qwen	contract	DEMOTE	0	0	1	0	19
Qwen	contract	ESCALATE	0	0	0	7	13
Flash	raw-strict	ALLOW	59	0	0	0	1
Flash	raw-strict	HOLD	0	17	0	0	3
Flash	raw-strict	DEMOTE	0	0	17	0	3
Flash	raw-strict	ESCALATE	0	0	0	17	3
Flash	contract	ALLOW	59	0	0	0	1
Flash	contract	HOLD	0	17	0	0	3
Flash	contract	DEMOTE	0	0	15	0	5
Flash	contract	ESCALATE	0	0	0	17	3